

הטכניון

מכון טכנולוגי לישראל

IoT project course

2360333

Smart Display
group 5

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Short description

A smart display that presents useful personal information such as calendar events, time, and other relevant data. The user can interact with the display through a graphical interface and send information or commands from a smartphone using voice input with Gemini integration.

User types

- User

User stories

#	Story Name	As a...	I want...	So that...
1	Voice Input	User	To add and remove items on my lists by speaking into my phone, in English or Hebrew	I can quickly add information to the display without manually typing it
2	Calendar Integration	User	To see my upcoming Google Calendar events on the display	I can keep track of my schedule without checking my phone
3	Graphical User Interface	User	To use the display through a clear touch interface with buttons	I can easily move between the display's features
4	Time and Date	User	To see the current time and date, with a greeting for the time of day	I can quickly check them from the display
5	Weather Information	User	To see the current weather and the forecast for the next days in my city	I can plan my day accordingly
6	Display Customization	User	To choose the look of the display: dark or light theme, clock format and my weather city	The display fits my taste and the place I live
7	Wi-Fi Setup	User	To connect the display to my Wi-Fi from my phone, and to choose a new Wi-Fi the same way if the old one is gone	I can start using its online features without reprogramming it or any technical configuration
8	Connection Status	User	To see on every screen whether the display is connected to the Wi-Fi	I understand why online information is not updating
9	Offline Mode	User	To keep using the display during network outages, with the last weather and events still shown and marked when they are old	The display remains useful when the internet connection is unstable
10	Brightness Control	User	To control the screen brightness with a slider	The display is comfortable to use in different lighting conditions
11	Settings	User	To change the display's settings in one Settings screen, grouped into sections	I can adapt it to my preferences
12a	Shopping List	User	To keep a shopping list on the display and tick items off when I buy them	I can track what I need to buy directly on the display
12b	To-Do List	User	To keep a separate to-do list and switch between my two lists	I can organize tasks the same way as my shopping
13.a	Command Understood	User	To see in the app what my command was understood as, before it reaches the display	I can check that my command was understood correctly
13.b	Command Result	User	To see the result of my command in the app and on the display, without being pulled away from what I am doing on the display	I know that my request was successfully processed, do not repeat it unnecessarily, and am not interrupted

14	Idle / Power Saving Mode	User	To have the display dim or turn off when no one is using it, and wake up when I touch it	I can save power while keeping the display quickly accessible when needed
15	World Clock	User	To see the time in several cities around the world	I can easily know the time for family, friends or colleagues abroad
16	Hebrew Support	User	To see list items and events written in Hebrew displayed correctly	I can use the display in my own language
17	Home Screen Layout	User	To choose which widgets the home screen shows and which one is big	The home screen shows what matters to me, in the size I need
18	Setup Without Code	Admin	To set up the voice commands and the calendar from my phone	I can set up or change the display's accounts without editing code or uploading new firmware
19	Commands While the Display Is Offline	User	To send commands even when the display is off or offline, see that it is offline, and have the commands applied when it is back	No command is lost and I don't have to wait or send it again
20	Undo a Command	User	To undo my last command with one tap in the app	I can quickly fix a mistake, such as an item added to the wrong list
21	Clear Answers to Unclear Commands	User	To be told what to say instead when my command is unclear, without anything being changed	Wrong items never end up on my lists, and I know how to say the command correctly
22	List Limits	User	To be told when an item can't be added because the list is full or the name is too long	I never lose an item without knowing
23	Synced Lists on Display and Phone	User	To change my lists both on the display and from my phone, even when one of them is offline, without losing changes	My lists are the same everywhere
24	Device Status	Admin	To see in one place whether the display is connected, whether voice commands work, and why it last restarted	I can find and fix a problem without connecting the display to a computer
25	Automatic Recovery	Admin	To have the display recover by itself from problems, and keep my lists and settings after a power cut	The display keeps working without my attention
26	Protected Access	Admin	To have the display's setup network protected by a password, and the app limited in how many commands can be sent	Strangers can't change my lists or overload the display
27	Helpful Messages	User	To see a short message that says what is wrong and what to do whenever a screen is empty or something isn't working	I never face an empty or confusing screen
28	See My Lists in the App	User	To see both of my lists on my phone	I can check my shopping list in the shop, away from the display
29	Type a Command	User	To type a command instead of speaking	I can use the lists in a noisy or quiet place, or without the microphone
30	Reset the Display	Admin	To erase everything saved on the display with one button, with a second tap to confirm	I can give the display to someone else or move it to a new home safely
31	Clear a List	User	To empty a whole list at once on the display, with a second tap to confirm	I can start a fresh list after shopping without removing each item

Note on Story 1 (Voice Input) - Google Assistant integration research:

Investigated Google Assistant integration for free-form custom commands (e.g. "add milk to my list"). Google Assistant's Conversational Actions, the only path that supported this kind of custom, free-form command, was deprecated by Google in June 2023 and is no longer available to developers. Standard Smart Home / App Actions only support predefined device traits (on/off, brightness) and cannot parse arbitrary custom commands. Fallback architecture adopted instead: a phone app records the voice command and sends it to the Gemini API, which returns structured JSON (e.g. {action, item, list}); this JSON is then passed to the ESP32 via a communication broker (Firebase Realtime Database or Adafruit IO / MQTT). See references 7, 8, 9 below.

Completion Status

all stories above are implemented in the final version. Changes from the original plan:

1. story 5 now has the option to choose which city's weather will be displayed (from 5 pre-selected cities).
2. Story 6 now covers the display's appearance (theme, clock format, weather city) instead of choosing which information is shown.
3. Story 11 has no language setting (Hebrew text is supported instead, story 16).
4. Story 13.b shows the interpreted action instead of the transcription (instructor feedback).
5. Story 14 wakes on touch only (the board has no motion sensor).
6. Stories 15 and 16 were added during development.
7. Stories 17 to 31 were added after mentor meeting 4 (18 October 2026): home screen customization (lecturer's suggestion), setup without code, voice commands while the display is offline, and the robustness and user-experience review.

Links to references, projects or tutorials that are related to your user stories

#	Relevance to our project (feature reference, technical reference)	Related to user stories	link
1	Existing ESP32 smart display project that displays Google Calendar events and weather information	2,5	ESP32 E-Ink Family Calendar project
2	Official Google Calendar API documentation showing how calendar events can be retrieved programmatically	2	Google Calendar API - Events List
3	Official Gemini documentation showing that voice/audio input can be processed in real time, suitable for sending voice commands from a smartphone	1,13.a,13.b,19,21	Gemini Live API / Live Transcription
4	Tutorial demonstrating a graphical touchscreen interface on ESP32, including buttons, sliders and touch input	4,6,8,10,11,12a,12b,17	ESP32 + LVGL TFT Touchscreen tutorial
5	Example showing communication/control of an ESP32 from a smartphone through Wi-Fi	1,7,18	ESP32 Web Server controlled from smartphone
6	Official Espressif documentation confirming ESP32 support for LCD displays and GUI libraries such as LVGL	6,8,10,11,12a,12b,17	Espressif LCD / GUI documentation
7	Firebase Realtime Database - REST API reference for ESP32 read/write, used as the communication broker for voice command JSON	1,13.a,13.b,19,20,23	Firebase REST API documentation
8	Adafruit IO - MQTT feeds reference, alternative communication broker for pushing voice command JSON to the ESP32 in real time	1,13.a,13.b	Adafruit IO MQTT documentation
9	Google's official notice on the deprecation of Conversational Actions (June 13, 2023), confirming direct free-form Assistant integration is not available	1	Conversational Actions sunset overview
10	WiFiManager library: phone setup page for the WiFi, extended with our own fields for the Firebase and Google Calendar settings	7,18	WiFiManager (github.com/tzapu/WiFiManager)
11	Gemini API JSON output mode, used to turn the recorded command into {transcript, action, item, list} that our server then checks by fixed rules	1,13.b,21	Gemini API - Structured output (JSON mode)
12	ESP-IDF task watchdog, used to restart the display automatically if it freezes	25	ESP-IDF Task Watchdog Timer documentation

List of hardware we think we will need for the project

Recommended to look at the parts catalog to see what is available

#	Part function	Why we need it (optional)
1	ESP32-2432S028R "Cheap Yellow Display" (ESP32 + 2.8" touchscreen), 1 unit	Runs the whole display: screen, touch, WiFi and all features on one board
2	Micro-USB cable + 5 V / 1 A USB power adapter	Power and programming the board